

# Harsath Kaliappa Gounder Thangavel

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## Education

### York University

Bachelor of Science (BSc) in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

## Professional Experience

### Software Engineer

Privex Inc.

October 2023 – January 2024

London, United Kingdom

- Developed an AI-driven support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, reducing staff workload by handling common inquiries.
- Implemented REST APIs to interface with the LLM and integrated Discord APIs for customer queries and staff data updates.
- Built data ingestion and management workflows to process context data from website documentation, bulk files, support tickets, and Discord announcements.
- Built and maintained infrastructure using MySQL for chunked data storage and ChromaDB for embedding retrieval.
- Containerized the Flask API with Docker for development and production deployments.
- Created systemd service files and monitoring scripts for system reliability.
- Tech Stack:** Python (Flask, ChromaDB, Langchain), MySQL, Linux

### Software Engineer

CogniPet AG

April 2019 – December 2019

Zürich, Switzerland

- Developed Pet Breed Detection system using Xception neural network and transfer learning, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet's Android and iOS apps using AWS EC2.
- Built data management workflows for user-uploaded pet images in AWS S3 with GDPR compliance.
- Explored video-based techniques for pet facial recognition through multi-frame feature extraction analysis.
- Tech Stack:** Python (Flask, Keras), AWS (EC2, S3)

### Founder and Full-Stack Web Developer

TaffyHub.com

August 2017 – July 2019

Coimbatore, India

- Developed and launched school social networking platform, reaching 500+ registered users and 150+ daily active users.
- Implemented news feed, friend recommender, content sharing, engagement system, and premium membership features.
- Built backend with PHP and MySQL, frontend with JavaScript, jQuery, and Bootstrap, using Memcached for query optimization.
- Self-hosted the platform on Linux infrastructure, implementing reverse proxies, automated backups, and system maintenance as part of DevOps workflow.
- Generated \$100/month revenue through Google AdSense and premium memberships.
- Tech Stack:** PHP, JavaScript, MySQL, Memcached, Linux (rsync, cron)

## Projects

### Jix Interpreter | C, CMake, GitHub Actions

- Developed an interpreter for a dynamically typed programming language in C.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.
- Tested using automated tests with GitHub Actions CI/CD pipelines.

### Prism Coverage | Java, Maven, JUnit, GitHub Actions

- Developed a Java test code coverage engine to measure test suite coverage against control flow and data-flow criteria.
- Implemented automated color-coded HTML report generation highlighting covered and uncovered sections of source code.

## Technical Skills

**Languages:** Python, Java, JavaScript, TypeScript, C++, C, Bash, HTML/CSS  
**Technologies:** Linux, Docker, SQL (MySQL, Postgres), NoSQL (MongoDB, Redis, Neo4j), AWS  
**Frameworks:** Node.js, React, Flask, Django, Langchain  
**Tools:** Git, CI/CD (Github Actions, Jenkins), CMake, Maven